

```

root@server0 Desktop
File Edit View Search Terminal Help
[root@server0 Desktop]# ipa-client-install --force-ntpd
DNS discovery failed to determine your DNS domain
Provide the domain name of your IPA server (ex: example.com): example.com
Provide your IPA server name (ex: ipa.example.com): ipa.example.com
Skip ipa.example.com: cannot verify if this is an IPA server
Failed to verify that ipa.example.com is an IPA Server.
This may mean that the remote server is not up or is not reachable due to net
work or firewall settings.
Please make sure the following ports are opened in the firewall settings:
TCP: 80, 88, 389
UDP: 88 (at least one of TCP/UDP ports 88 has to be open)
Also note that following ports are necessary for ipa-client working properly
after enrollment:
TCP: 464
UDP: 464, 123 (if NTP enabled)

```

Figure 3: Creation of an IPA client

Using Authconfig-gtk

To use Authconfig-gtk to configure a system for LDAP + Kerberos, follow the steps given below.

- Install all the necessary packages as shown below:

```
# yum -y install authconfig-gtk sssd krb5-workstation
```

- Launch authconfig-gtk, either from the command line or from *Applications > sundry > Authentication*. Enter the root password when prompted.
- On the identity and authentication tab, select *LDAP* from the user account database drop-down menu. Fill in the LDAP search base DN and LDAP server fields.
- If the LDAP server supports TLS, check the 'Use TLS to encrypt connections' box and use the 'Download CA Certificate' button to download the CA certificate.
- From the 'Authentication Method' drop down menu, select the Kerberos password and fill out the realm, KDCs and admin server fields. The last two fields are not available if the 'Use DNS to locate KDCs for Realms' option is checked.
- If central home directories are not available, users can create directories on first logging in by checking the 'Create home directories' on the first login box on the 'Advanced Options' tab.
- Click the 'Apply' button to save and activate the changes. This will write all relevant configuration files and restart the sssd service.

Testing a configuration

To test the LDAP + Kerberos configuration, an administrator can attempt to log into the system (over SSH) using the credentials of one of the network users. In addition, the *getent* command:

```
getent password <USERNAME>
```

...can be used to retrieve information about a network user.

Important: In the default configuration, sssd will not enumerate the network users when no user name is

specified to the *getent* command. This is done to keep the graphical login screen uncluttered to save valuable network resources and time.

Attaching a system to an IPA server

Red Hat provides an integrated solution for configuring LDAP and Kerberos – the IPA (identity, policy and auditing) Server. This provides LDAP and Kerberos, combined, with a suite of both command line and Web based administration tools. Apart from user and authentication information, IPA Server can centralise sudo rules, SSH public keys, SSH host keys, TLS certificates, automounter maps, and much more.

Using ipa-client

A RHEL 7 system can be configured to use IPA Server with the Authconfig suite of tools, but a specialised tool also exists: *ipa-client-install*. This command can be installed from the ipa-client package, which pulls in all dependencies (such as sssd).

One of the benefits of using *ipa-client-install* is that it can retrieve almost all necessary information from DNS (when configured either by IPA Server or manually by an administrator), as well as create host entries and more on the IPA Server. This allows an IPA Server administrator to set access policies for that host, create service principles (e.g., for NFSv4 exports), and more.

When *ipa-client-install* is run without any arguments, it will first attempt to retrieve information about the IPA Server configured for its DNS domains from the DNS. If that fails, it will prompt the administrator for necessary information, such as the domain name of IPA Server. Unless another account has been created for this, the default IPA Server administrator account (admin) can be used.

Figure 3 gives an example of a driven configuration.

It is possible to specify all the required information as command line arguments, allowing for unattended setups as part of an initial system configuration, for example, from Kickstart.

Joining a system to Active Directory

RHEL 7 has many ways of joining a system to Active Directory. Administrators can choose to install the Samba-Winbind package and configure Winbind through the Authconfig family of tools, or the administrator can install both sssd and realmd packages and use sssd and realm commands.

Note 1: The realm command can also be used to join Kerberos realms or IPA Server domains, but the final configuration will be slightly different; for example, users will have @domain appended to their user names. Ipa-client-install is the preferred method to join IPA domains.